

Nicholas Goh

AI Full Stack Engineer

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Professional Experience

Knovel Engineering

07/2025 – Present

Software Engineer

- Designed bespoke testing strategies for customer agentic systems
- Evaluated security, reproducibility, explainability, and guardrail robustness across client systems
- Dogfooding AI Assurance Platform 🔗
- Analyzed customer's fine-tuned model against RAG system
- Enhanced RAG to support complex tables and flowcharts.

Klass Engineering and Solutions

08/2023 – 05/2025

Software Engineer (AI Engineering)

- Designed system and implemented an app to showcase LLM Orchestration Capabilities. This opened opportunities for future work with AI Agents delegating and executing tasks
- Modularizing UI into a reusable base Chatbot framework to reduce technical debt and streamline development, cutting UI development and integration time for future teams.
- Architected and developed an in-house centralized model weights caching, to address scaling and MLOps challenges. This resulted in 3% (2TB/70TB) of disk space savings and 5 days of time savings per developer
- Analyzed third party codebase to identify and debug critical RAM and VRAM leak, resulting in 100% improvement of efficiency

Projects

FastAPI MCP LangGraph Template 🔗

04/2025 – 07/2025

- Evaluated trade-offs between NoSQL, Milvus, PostgreSQL and PGVector extension, accessing the additional complexity required to implement cross database consistency
- Evaluated trade-offs between using AWS Lambda + Amplify and EC2 for deployment, opting for EC2 to simplify local and server testing for both backend and UI
- Leveraged IaC with Terraform to automate provisioning of AWS EC2, ECR, and security policies, enabling cost-effective and reproducible setup and teardown of cloud resources
- Optimized document ingestion and vectorization on a small EC2 instance, using a rolling window approach to avoid memory constraints and preserve context between chunks

Customer Service Automation 🔗

02/2025 – 03/2025

- Built an AI-powered system that automatically handles multi-part queries, significantly reducing manual effort, improving response times, and increasing operational efficiency in customer interactions
- Implemented automated error handling and conflict resolution, ensuring more reliable booking processes, minimizing disruptions, and enhancing overall user experience
- Introduced LLM-based testing and LangSmith tracing to ensure high-quality, consistent outputs from AI agents, significantly reducing troubleshooting time and improving overall system stability and performance

Skills

Cloud & Devops: GCP, Grafana, Prometheus, GH Actions | **Databases:** Cassandra, Redis, Sqlite |
Languages & Frameworks: Python, Fastapi, Langgraph, Langchain, Typescript, React, CPP